

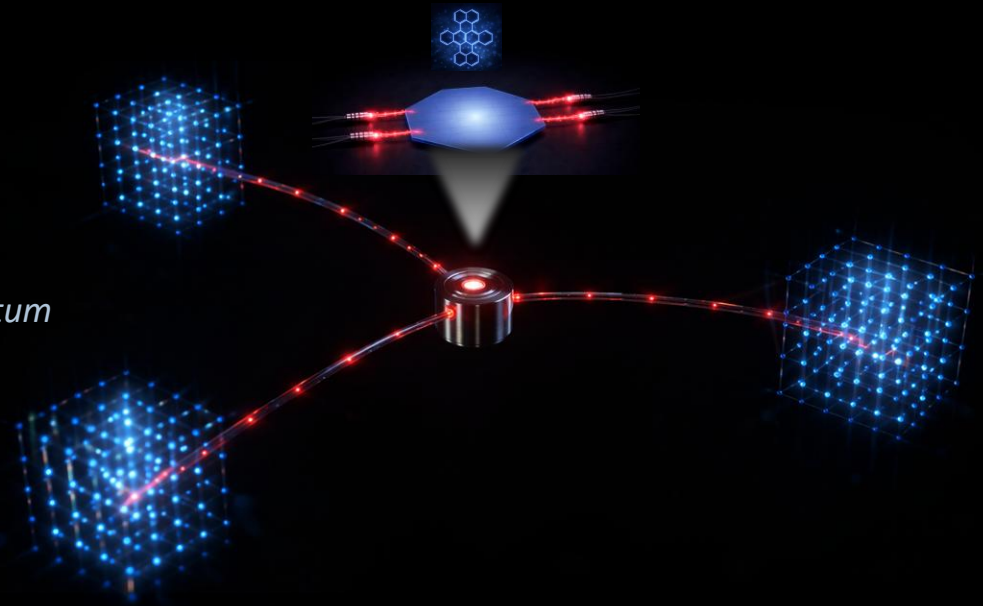


QCLUSTER PHOTONICS

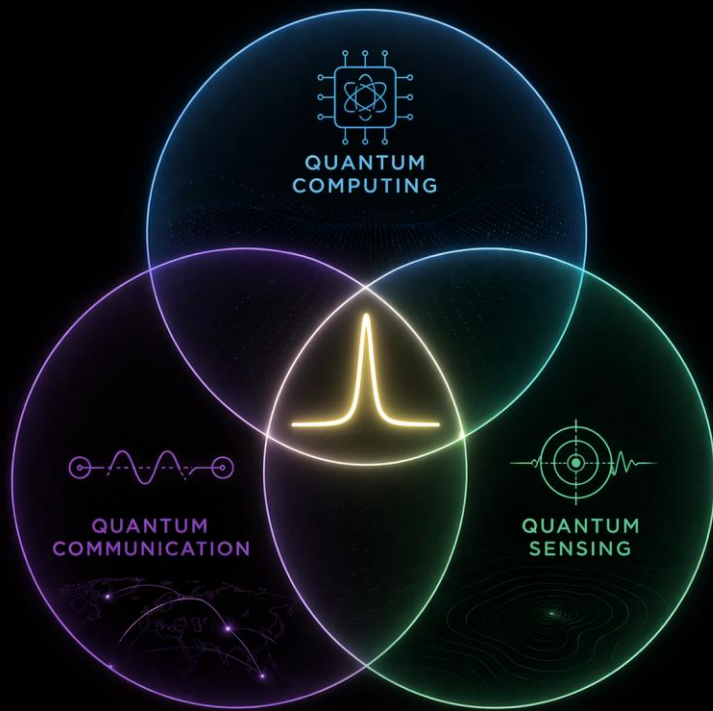
The interconnect layer for neutral atom quantum computers

PRE-SEED INVESTMENT ROUND

Dr. Christian M. Lange, Founder & CEO
Dr. Jonathan D. Hood, Founder & CTO



A Bottleneck in a Trillion-Dollar Opportunity



Interconnects are the missing piece

Quantum processors are scaling rapidly, but quantum interconnect technology is lagging far behind. In 3-5 years, the primary bottleneck for quantum computation will be the ability to link processors together into a quantum datacenter.

“Networking quantum computers is the critical connective tissue.”

— IBM Quantum

McKinsey Quantum Technology Monitor 2026;
Grand View Research; Markets and Markets; IBM Quantum (2025).

Rethinking the networking layer from the ground up

We believe a truly scalable networking layer requires two elements

1 An external photon source

A bright single-photon supply for remote entanglement.

Current approaches require trapped atoms to simultaneously be coherent memories and to supply pure photons. An external photon source unlocks entanglement protocols with lower sensitivities and greater scalability.

2 Qubit–nanophotonic interfaces

Efficient, scalable coupling between qubits and nanophotonic circuits

Current approaches rely on free-space cavities, sacrificing the scalability of nanophotonics



Our technology

We are tackling both requirements of quantum networking head-on

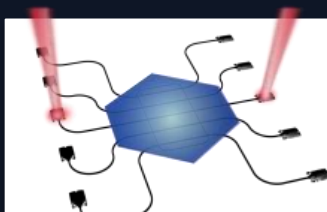
1 Deterministic photons for atoms

We engineered a photon source for compatibility with trapped atoms

35 MHz linewidth

Aligned to the Rb D2 transition

Nanophotonic integration for scalability



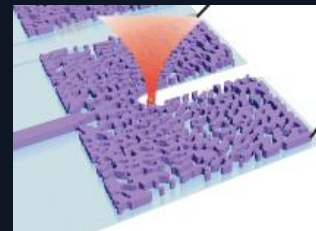
Patent pending

2 Nanophotonic interfaces for atoms

We are engineering nanophotonics around the constraints of laser-cooled atoms

Scattering-induced heating

Casimir-Polder forces



Product direction: An entanglement router

>95%

Fidelity

100 kHz

Per-atom entanglement rate

100×

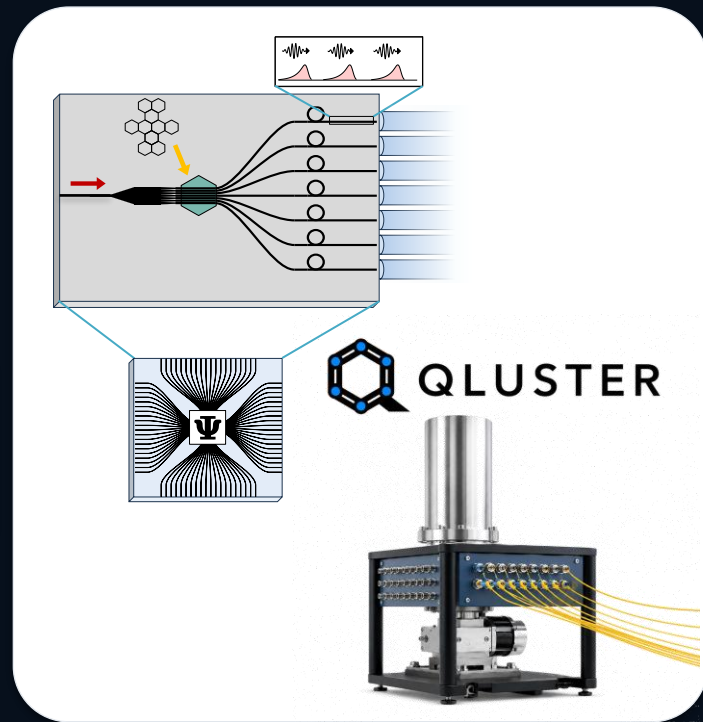
Output channels

Integrated SNSPD
detectors, Bell-state
analyzers, and phase
stabilization

Single photon heralding
remote entanglement
generation scheme with
high fidelity and rates

Quantum frequency
conversion of >70% to
target all neutral and ion
transitions

\$250k base price + upgrades, service, and installation



We have the only photon emitter technology that is both deterministic and compatible with atomic qubits.

Other sources are too broadband, or wavelength mismatched to be compatible with atoms.

Market sizing

TAM \$10-\$50B

Longterm interconnect infrastructure for quantum datacenters, QKD networks, photonic QC, and sensing.

SAM \$1B

Thousands of interconnects needed for 1st-gen quantum datacenters.

SOM \$100M

Rb-compatible systems, research labs, and quantum-network testbeds. Path to 500 units at \$200k ASP.

Customer markets

1 Atom & ion quantum computing

Photonic links for remote entanglement between modular QPUs.
QuEra · Atom Computing · Infleqion · Pasqal · IonQ · Quantinuum

2

Quantum networks, repeaters & memories

Narrowband photons for storage, retrieval, entanglement swapping & repeater nodes.
Qunnect · Nu Quantum · NanoQT · Quantum Bridge · memQ

3

Photonic QC, QKD & sensing

Deterministic photons for optical qubits, secure links & detector calibration.
PsiQuantum · Xanadu · Quandela · ORCA · ID Quantique · Quantum Opus

Customer traction

Customer pull from quantum networks, Rb memories, and atom-based processors.

NanoQT · Aki Goban

LOI offered for 780 nm photon source

Bristol QET Labs · Alex Clark

Interested in 780 nm photons for a quantum network based on Rb vapor.

Purdue · Shengwang Du

Validation path using QCluster photons for storage and retrieval in Rb vapor.

Quantum Opus

Exploring source + detector partnership; shared cryogenic-photonics infrastructure.

WHY ADOPTION IS EASY

Customers already run SNSPD photon detectors. We bring the matching photon source.

Same cryogenic ecosystem, same photonic stack — QCluster is designed for infrastructure that customers already know how to run.

Roadmap

Q4 2026 – Q4 2028
Pre-Seed

- Founder transitions full-time
- First 2 hires
- Build photon-source prototype
- Demonstrate first atom-nanophotonic coupling
- Secure paid pilot from NanoQT for photon source
- Expand IP portfolio

Q4 2028 – Q4 2029
Seed

- Demonstrate 100 kHz REG with nanophotonic-coupled atoms and external photon source
- Quantum networking module co-development with strategic partner

Q1 2030 →
Series A

- First product shipped
- Revenue: \$2M → \$5M → \$10M / yr
- Expand platform: Rb → photonic QC + QKD + sensing

Q4 2028

Key inflection point

Photon source prototype complete · Atom-nanophotonic proof-of-principle → unlocks Seed round

Q1 2031

Commercial launch

First product shipped · \$2M Year-1 revenue → path to scale

QLUSTER Founders



Dr. Christian M. Lange

Co-founder & CEO

PhD Purdue 2025 — built the lab from empty bench to breakthrough in 4 years.

First author on the peer-reviewed papers underpinning QLuster's patent. Hands-on expert in **nanofabrication, crystal synthesis, and device characterization**. Every breakthrough in our IP stack was made by him at the bench.



Prof. Jonathan Hood

Co-founder & CTO

Asst. Professor, Chemistry & Physics, Purdue — PhD Caltech (Kimble) · Postdoc Harvard.

Runs labs in both solid-state quantum emitters and ultracold atom physics — one of the only faculty in the world who bridges these two fields. Publications in Science, Nature Physics, PRL, ACS Nano; 2,800+ citations; \$3M+ in research funding.

THE ASK

\$3M

Pre-Seed Round

Terms negotiable

IN 24 MONTHS, DELIVER

- ✓ Photon-source prototype with 10% molecule-to-fiber coupling efficiency
- ✓ Atom-nanophotonic coupler proof-of-principle
- ✓ Paid pilot with NanoQT for photon source
- ✓ Expand IP portfolio to control molecular photon source and atom-nanophotonic coupler

HOW THE FUNDS ARE DEPLOYED

Personnel \$1M

2 hires at \$200k/year each; founder \$100k/year

Atom-trapping experiment \$500k

Vacuum, objective, lasers, optics, stages and photon-detecting camera

Photon source optics and synthesis \$500k

Cryostat, CW laser, EOMs, photon detectors, time tagger, DBT synthesis

Nanofabrication + simulation \$200k

\$20k materials + \$100k fabrication usage + \$20k workstation + \$10k simulation credits + \$50k LLM usage

Operations \$150k

Purdue lab space and license royalty

Base program \$2.35M

Contingency and headroom \$650k

Total allocation \$3M

WHAT COMES NEXT

Seed Round — 2028

Series A — 2030

Revenue Ramp — 2031–2033

\$2M → \$5M → \$10M / year

Let's Build the InfiniBand of the Quantum Age

QLuster Photonics | Pre-Seed Investment Round



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